Test Case Document (updated after Sprints 1 and 2)

Project: Railway Reservation System (RRS), Team 13 Jackfruit Phase-1
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Environment: Python 3.11.9, Windows
Replaces: the test case list in STP v1.0, section 13

Result summary

STP test cases	Passing (automated)	Manual only	Failing
27	27	0	0

Test cases

TC	Requirement	What must be true	Level	Automated tests	Result	Linked by
TC-Auth-01	RRS-F-001	Valid registration creates an account; duplicate email is rejected	Unit	2	Pass	tag
TC-Auth-02	RRS-F-002	Correct credentials grant access to the dashboard	Unit	2	Pass	tag
TC-Auth-03	RRS-F-003	After 5 failed logins further attempts are blocked temporarily	Unit	8	Pass	tag
TC-Search-01	RRS-F-010	Matching trains are returned with class and timing	Unit	1	Pass	tag
TC-Search-02	RRS-F-011	'No trains found' message for an unmatched route or date	Unit	1	Pass	tag
TC-Book-01	RRS-F-020	Availability count matches the allocation state	Unit	5	Pass	file
TC-Book-02	RRS-F-021	Provisional booking is created and held for a limited time	Unit	4	Pass	file
TC-Book-03	RRS-F-022	Missing or invalid booking fields are flagged	Unit	4	Pass	file
TC-Pay-01	RRS-F-030	Booking becomes Confirmed only after a Success payment	Integration	8	Pass	file
TC-Pay-02	RRS-F-031	Failed or timed-out payment releases the seat	Integration	17	Pass	tag
TC-PNR-01	RRS-F-040	A confirmed booking returns a valid PNR; lookup shows the right status	Unit	3	Pass	tag
TC-Cancel-01	RRS-F-041	Confirmed cancellation sets Cancelled and releases the seat	Unit	2	Pass	tag
TC-Cancel-02	RRS-F-042	Refund amount matches the cancellation policy	Unit	4	Pass	tag
TC-Alloc-01	RRS-F-050	Assigned status matches current capacity	Unit	3	Pass	tag
TC-Alloc-02	RRS-F-051	A cancellation promotes the earliest eligible waiting booking	Integration	16	Pass	tag

TC-Admin-01	RRS-F-060	Schedule changes are reflected in search results	Unit	24	Pass	tag
TC-Admin-02	RRS-F-061	Report figures match the underlying booking data	Unit	21	Pass	tag
TC-Perf-01	RRS-NF-001	90th percentile transaction time is 5 seconds or less	Performance	2	Pass	tag
TC-Sec-01	RRS-NF-003	No sensitive data is stored or exposed in plaintext	Security	7	Pass	tag
TC-Ops-01	RRS-NF-004	Timestamped audit logs can be retrieved for sample bookings	Integration	5	Pass	tag
TC-UX-01	RRS-NF-005	Labels, defaults and contrast pass; walkthrough done without help	Usability	10	Pass (automated); manual walkthrough pending	tag
TC-Sec-02	PRJ-SR-001	Connections without TLS are rejected; TLS 1.2 or newer only	Security	7	Pass	tag
TC-Sec-03	PRJ-SR-002	Passwords are stored only as salted hashes	Security	1	Pass	tag
TC-Sec-04	PRJ-SR-003	Direct API calls to admin endpoints are rejected for passengers	Security	21	Pass	tag
TC-Sec-05	PRJ-SR-004	Session ends after the idle timeout	Security	7	Pass	tag
TC-Sec-06	PRJ-SR-005	No raw card-like data is stored; only reference and status	Security	1	Pass	file
TC-Sec-07	PRJ-SR-006	An audit entry exists for every admin action	Security	11	Pass	tag

`Linked by: tag` means the test names the TC id in its docstring or a comment. `file` means the whole test file was mapped to the TC; check it. Add the TC id to a test's docstring to link it exactly.

Changes since the submitted Test Plan (STP v1.0)

- Database: the build uses Supabase PostgreSQL, not MySQL as written in the SRS, SAD and STP. Repository code is tested with a fake cursor; SQL is exercised in system validation.
- Mock payment: the outcome (Success, Failed, Timeout) is chosen on the booking page for demonstration, which is how the planned Success/Failure test scenarios are driven.
- Added in Sprint 2: lockout (TC-Auth-03), idle timeout (TC-Sec-05), admin-only checks (TC-Sec-04), audit trail (TC-Sec-07, TC-Ops-01), waitlist/RAC promotion (TC-Alloc-02), train and schedule admin (TC-Admin-01), TLS (TC-Sec-02), health and uptime (RRS-NF-002), usability checks (TC-UX-01).
- TC-Sec-02 runs a real TLS handshake and checks that TLS 1.1 is refused, instead of only reading config.
- TC-UX-01 is split into an automated part (labels, contrast, defaults) and a manual walkthrough (docs/TC-UX-01_usability_walkthrough.md).
- RRS-NF-002 (99% availability) is evidenced by the uptime probe log, not by a pytest test.